Online Resource 1 for "When Does Relational Structure Help Healthcare Fraud Detection? A Controlled Benchmark of Heterogeneous Graph Neural Networks and Graph-Augmented Tabular Models", *Health Information Science and Systems*

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Abstract

This Online Resource accompanies the article named in the title. It contains material summarised but not shown in full there: the selected hyperparameters, the two-stage routing evaluation, the ring-volume dose-response, the confirmatory test of the hybrid result, three robustness checks (feature shift at reduced ring volume, training-seed replication and per-regime tuning), results for other ring shapes and sizes, the cost of training and scoring, single-factor ablations of HGT, which inputs the trained models use, figures of the mechanism ablations and of new versus recurring rings, repeated cross-validation and per-rotation tuning on the public data, a re-analysis of a small fixed benchmark, a wide hyperparameter search, the closure-column experiment and the analysis of published group-fraud flags. Every number is generated from the run records in the article's code repository.

S1 Selected hyperparameters

Table S1 lists the configurations selected with the equal tuning budget.

Table S1 Selected hyperparameters (equal budget per method; hybrids inherit their base configuration; methods sharing a search space were offered the same candidate configurations). Synthetic: tuned on development worlds; public data: tuned on a separate split. The configurations re-tuned for the low-volume stress test are in the repository (`results/tuning_lowvolume`).

Method	Synthetic benchmark	Public data
LR	C=100	C=1
RF	max depth=16; max features=sqrt; min samples leaf=3; n estimators=300	max depth=16; max features=0.3; min samples leaf=10; n estimators=300
GB (C)	l2=0; learning rate=0.03; max iter=200; max leaf nodes=15; min samples leaf=10	–
GB (C+A / P)	l2=1; learning rate=0.1; max iter=200; max leaf nodes=63; min samples leaf=50	l2=0; learning rate=0.03; max iter=200; max leaf nodes=15; min samples leaf=50
GB (C+A+G / P+G)	l2=1; learning rate=0.1; max iter=400; max leaf nodes=15; min samples leaf=20	l2=0; learning rate=0.03; max iter=200; max leaf nodes=15; min samples leaf=50
GB (P+G+L)	–	l2=0; learning rate=0.03; max iter=200; max leaf nodes=15; min samples leaf=50
MLP	dropout=0.3; hidden=128; layers=2; lr=0.001; weight decay=1e-05	dropout=0.3; hidden=128; layers=2; lr=0.001; weight decay=1e-05
SAGE	aggr=sum; dropout=0.1; hidden=32; layers=3; lr=0.003; weight decay=0.001	aggr=mean; dropout=0.3; hidden=64; layers=3; lr=0.003; weight decay=0.0001
RGCN	aggr=sum; dropout=0.3; hidden=64; layers=2; lr=0.003; weight decay=0.001	aggr=sum; dropout=0.3; hidden=64; layers=2; lr=0.01; weight decay=0.001
HAN	dropout=0.3; heads=4; hidden=32; layers=3; lr=0.01; weight decay=1e-05	dropout=0.1; heads=2; hidden=64; layers=2; lr=0.003; weight decay=0.0001
Simple-HGN	dropout=0.3; heads=4; hidden=64; layers=3; lr=0.003; weight decay=0.0001	dropout=0.1; heads=4; hidden=64; layers=2; lr=0.01; weight decay=0.001
HGT	dropout=0.3; heads=4; hidden=32; layers=3; lr=0.01; weight decay=1e-05	dropout=0.1; heads=4; hidden=64; layers=2; lr=0.01; weight decay=0.001
BWGNN	dropout=0.3; hidden=32; lr=0.001; order=2; weight decay=0.0001	dropout=0.3; hidden=64; lr=0.003; order=3; weight decay=0.001
CARE-GNN	dropout=0.3; hidden=64; layers=2; lr=0.001; weight decay=0.0001	dropout=0.1; hidden=64; layers=2; lr=0.001; weight decay=0.0001

S2 Two-stage routing

S2.1 Method

A common deployment proposal is to score every claim with a cheap tabular model and pass only high-risk claims to a GNN. We evaluate this with GB-CAG as the first stage and HGT or SAGE as the second. A claim is escalated if its GB-CAG score is at least the $(1 - e)$ quantile of the GB-CAG scores in the validation period. The escalation fraction $e \in \{0.05, 0.1, 0.2, 0.3, 0.5, 1\}$ is chosen by validation AP, so test claims never influence it. Escalated claims rank above all other claims and are ordered either by the GNN score alone (*re-rank*: GB-CAG acts as a filter) or by the mean of the two models' validation percentiles (*fused*); non-escalated claims keep their GB-CAG order. With $e = 1$, re-ranking reduces to the GNN alone and fusion to a rank-average ensemble of the two models, so both single models and the ensemble are among the candidates. A claim's routed score depends only on its own two scores and on validation statistics. The stages are the models of the main experiment, retrained with the same seeds; their test predictions reproduce the stored ones exactly for the GNNs and to within 3×10^{-8} for GB-CAG (float32 storage).

S2.2 Results

Table S2 and Fig. S1 evaluate two-stage routing (Section S2.1) on the same test claims as the single models. In the design usually proposed, GB-CAG filters and the GNN decides (re-rank). Validation mostly selected small escalation fractions (5% in 30 of 54 worlds for HGT), and the routed system matched GB-CAG (GB-CAG→HGT ΔAP $= -0.001$, 95% CI $[-0.006, 0.003]$; GB-CAG→SAGE $+0.001$). Escalating more claims to HGT lowered both ring AP (from 0.980 at 5% to 0.968 when every claim is escalated) and, more steeply, isolated AP (from 0.626 to 0.508; Fig. S1b–d), because HGT ranks both less well than GB-CAG. Ordering escalated claims by the mean of the two models' percentiles did better: GB-CAG→HGT gained $+0.003$ (Holm $p = 0.60$) and GB-CAG→SAGE $+0.006$ $[0.003, 0.010]$ (Holm $p = 0.042$, positive for all six seeds), on ring ($+0.006$) and isolated fraud ($+0.023$) alike. Validation, however, chose to fuse *every* claim in 22 (HGT) and 30 (SAGE) of the 54 worlds, and on average 55% and 70% of test claims were escalated: the gain comes from combining two different rankings, not from routing a subset of claims. The feature-level hybrids, which give the GNN the engineered features, had slightly higher ring AP ($+0.008$, not significant with the seed as unit) but lost 0.08–0.11 isolated AP and so matched GB-CAG overall. At the ring level all methods reached the ceiling: in all 54 worlds each test ring had at least one claim among the top 5% of claims, and precision in that budget was 0.989–0.998 for every method.

S3 Ring-volume dose-response

Table S3 gives ring AP for every method at four ring volumes (16, 8, 4 and 2 claims per ring per month) at $\delta = 0$, over seeds 1–6, whose legitimate claims, isolated fraud and ring members are identical across volumes. The hyperparameters are those of the main experiment; the re-tuned results at volume 4 are in Table 4 of the article. In

Table S2 Two-stage routing versus single models and feature-level hybrids, averaged over the 54 worlds (same test claims for every row). Routing escalates claims whose GB-CAG score exceeds a threshold set on validation data; the escalation fraction is chosen by validation AP. P@5%: precision among the top 5% of test claims. Esc.: mean share of test claims escalated. ΔAP: paired difference in AP from GB-CAG (± half-width of the 95% t-interval over the six generator seeds), Holm-adjusted p over the rows.

Method	AP	Ring AP	Isol. AP	P@5%	Esc.	ΔAP (p)
GB-CAG	0.947	0.979	0.626	0.996	–	–
HGT	0.929	0.968	0.507	0.989	–	−0.018 ± 0.013 (0.104)
SAGE	0.940	0.979	0.541	0.995	–	−0.007 ± 0.009 (0.597)
HGT+AG	0.945	0.986	0.514	0.996	–	−0.002 ± 0.006 (1.000)
SAGE+AG	0.947	0.987	0.548	0.996	–	0.000 ± 0.005 (1.000)
GB-CAG→HGT, re-rank	0.945	0.979	0.606	0.995	0.16	−0.001 ± 0.005 (1.000)
GB-CAG→HGT, fused	0.950	0.981	0.640	0.997	0.55	+0.003 ± 0.004 (0.597)
GB-CAG→SAGE, re-rank	0.948	0.983	0.595	0.997	0.24	+0.001 ± 0.004 (1.000)
GB-CAG→SAGE, fused	0.953	0.985	0.648	0.998	0.70	+0.006 ± 0.004 (0.042)

a test period a ring patient files on average 9.3, 5.4, 3.7 and 3.0 claims at the four volumes, against 1.6 for other patients (95th and 99th percentiles about 3.8 and 5.2), and a ring doctor 38, 26, 20 and 18 claims against 16. At 2 claims per month a ring files about one claim per member, rings make up 1.1% of the non-isolated claims, and no method is far above chance. Paired comparisons for every cell are in the repository (`results/tables/dose_response_comparisons.csv`).

S4 Confirmatory test of the hybrid result

In seeds 1–6, after re-tuning at volume 4, SAGE+AG had higher ring AP than GB-CAG for sparse rings ($s = 0.1$: 0.129 against 0.078, Holm $p = 0.035$), the only significant advantage over GB-CAG in the low-volume experiments. Because this comparison was singled out after the results had been seen, we tested it once more as fixed in an analysis plan committed before the runs started (`paper/hiss_review/analysis_plan_round2.md`): fresh generator seeds 7–18, $\delta = 0$, $s = 0.1$, volume 4, the re-tuned configurations, only GB-CAG and SAGE+AG, and one two-sided paired t-test at $\alpha = 0.05$. SAGE+AG had a mean ring AP of 0.129 against 0.104 for GB-CAG: a difference of +0.024 (95% CI $[0.005, 0.043]$, $p = 0.016$; Wilcoxon $p = 0.009$), positive in nine of twelve seeds, with a chance level of 0.021. By the pre-declared rule the advantage is confirmed; its size is about half the exploratory estimate.

That test used one training seed, the one used for tuning. A later plan, also committed before its runs (`paper/hiss_review/analysis_plan_round5.md`), repeated it with training seeds 1 and 2, with a two-sided paired t-test at $\alpha = 0.05$ on each world's mean over the three training seeds as the primary test. GB-CAG's scores were identical for every training seed in these worlds, so only SAGE+AG varied. Its mean ring AP was 0.136 against 0.104: a difference of +0.032 (95% CI $[0.012, 0.052]$, $p = 0.005$;

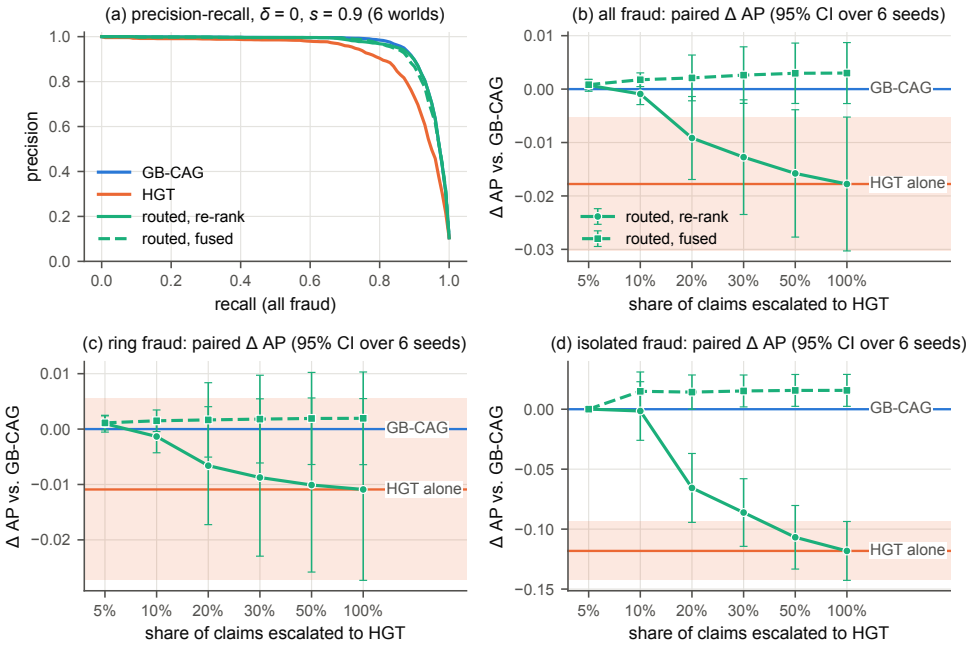


Fig. S1 Two-stage routing from GB-CAG to HGT. (a) Precision–recall curves for all fraud at $\delta = 0$, $s = 0.9$ (precision interpolated at fixed recall and averaged over six worlds; routed systems at the escalation fraction chosen on validation). (b)–(d) Paired difference in test AP from GB-CAG (zero line) against the share of claims escalated, averaged over the 54 worlds, with 95% CIs over the six seeds; orange line and band: HGT alone. The escalation fraction actually used is chosen on validation; the curves over all fractions are shown only to describe the trade-off.

Wilcoxon $p = 0.002$), positive in eleven of twelve seeds. By training seed the difference was +0.024, +0.028 and +0.044 ($p = 0.016$, 0.019 and 0.005; descriptive). By the pre-declared rule the advantage is robust to the training seed.

S5 Feature shift at reduced ring volume

In the article, ring volume was reduced to 4 claims per month only at $\delta = 0$ (Section 6.6). Here that test is repeated with ring claims also shifted, $\delta = 1$ and 2, as fixed in the plan for the third round of experiments before its runs (`paper/hiss_review/analysis_plan_round3.md`): seeds 1–6, all 14 methods, the hyperparameters of the main experiment, and legitimate claims, isolated fraud and ring members shared with the $\delta = 0$ and default-volume worlds of the same seed. Table S4 gives ring AP. Per-density p-values are Holm-adjusted within each method's three densities, pooled ones over methods with the seed as unit; every comparison is in the repository (`results/tables/lowvolume_shift_comparisons.csv`).

With a moderate shift ($\delta = 1$), coordination structure was again the main evidence for moderately and densely linked rings: GB-CAG exceeded GB-CA by +0.288 and +0.301 at $s = 0.5$ and 0.9 (Holm $p < 0.01$), against +0.039 and +0.048 at default

Table S3 Ring AP at $\delta = 0$ by ring volume (claims per ring per month; 16 is the default) and structural density s, mean over six worlds (seeds 1–6, which share their legitimate claims across volumes), hyperparameters of the main experiment. Chance level: 0.080 at 16, 0.042 at 8, 0.021 at 4, 0.011 at 2.

	$s = 0.1$				$s = 0.5$				$s = 0.9$			
Method / volume	16	8	4	2	16	8	4	2	16	8	4	2
GB-C	0.084	0.046	0.024	0.014	0.086	0.044	0.023	0.012	0.087	0.044	0.022	0.015
MLP-C	0.088	0.044	0.024	0.013	0.087	0.041	0.024	0.013	0.095	0.046	0.024	0.013
LR-CA	0.923	0.524	0.106	0.025	0.888	0.428	0.087	0.039	0.868	0.364	0.088	0.020
RF-CA	0.875	0.339	0.057	0.019	0.838	0.343	0.064	0.018	0.844	0.331	0.064	0.018
GB-CA	0.925	0.422	0.049	0.020	0.905	0.372	0.052	0.019	0.889	0.311	0.055	0.018
GB-CAG	0.930	0.550	0.066	0.018	0.983	0.879	0.409	0.024	0.990	0.923	0.570	0.019
SAGE	0.938	0.606	0.074	0.016	0.963	0.683	0.108	0.017	0.976	0.755	0.131	0.018
RGCN	0.933	0.546	0.066	0.018	0.953	0.637	0.112	0.018	0.954	0.686	0.142	0.018
HAN	0.908	0.469	0.062	0.014	0.906	0.507	0.065	0.018	0.921	0.520	0.058	0.014
Simple-HGN	0.915	0.561	0.074	0.014	0.960	0.692	0.133	0.015	0.957	0.716	0.151	0.013
HGT	0.910	0.544	0.046	0.016	0.948	0.649	0.079	0.017	0.949	0.689	0.098	0.016
BWGNN	0.904	0.487	0.065	0.017	0.918	0.487	0.076	0.018	0.924	0.469	0.079	0.016
CARE-GNN	0.882	0.407	0.057	0.014	0.863	0.318	0.051	0.017	0.868	0.351	0.053	0.013
SAGE+AG	0.973	0.659	0.103	0.020	0.985	0.883	0.379	0.032	0.983	0.914	0.566	0.025
HGT+AG	0.964	0.546	0.074	0.019	0.984	0.872	0.324	0.024	0.990	0.916	0.523	0.023
Overlap score	0.254	0.141	0.057	0.018	0.937	0.721	0.194	0.030	0.985	0.866	0.281	0.029

volume, and by +0.017 at $s = 0.1$ (not significant). With a strong shift ($\delta = 2$) the claims themselves carried most of the evidence and the structure gain was small (+0.036 pooled over densities, 95% CI $[-0.002, 0.074]$, Holm $p = 0.27$), as at default volume.

For sparse rings ($s = 0.1$), several methods outperformed GB-CAG, one of them without graph inputs. At $\delta = 1$, HAN exceeded it by +0.178 (Holm $p = 0.020$), SAGE by +0.139 ($p = 0.030$), SAGE+AG by +0.123 ($p = 0.042$) and logistic regression on claim and aggregate features (LR-CA) by +0.109 ($p = 0.022$); at $\delta = 2$, SAGE, HAN, RGCN, both hybrids and LR-CA exceeded it by +0.053 to +0.087 (Holm $p \leq 0.037$). Gradient boosting was comparatively weak in these cells (GB-CA 0.269 against 0.395 for LR-CA at $\delta = 1$), and the best pure GNNs were 0.03–0.07 above LR-CA (SAGE 0.425 and HAN 0.464 at $\delta = 1$; this comparison was not tested). Because a linear model without graph inputs also outperformed gradient boosting here, these advantages are better explained by how well each model learns a weak claim-level shift from about 200 training ring claims than by message passing detecting coordination, which added little for sparse rings.

For moderately and densely linked rings at $\delta = 1$, the pure GNNs were 0.11–0.29 below GB-CAG (significantly only for HGT, Holm $p < 0.01$, with six seeds) and SAGE+AG was 0.04–0.07 above it (Holm $p \geq 0.079$). Pooled over densities, no method had significantly higher ring AP than GB-CAG at either shift (largest: SAGE+AG, +0.079 $[0.013, 0.146]$ at $\delta = 1$, Holm $p = 0.20$, and +0.037 $[0.016, 0.057]$ at $\delta = 2$, $p = 0.055$), and HGT was significantly lower at $\delta = 1$ (-0.202, $p = 0.007$). On isolated

fraud the GNNs and hybrids were again below GB-CAG, by 0.02–0.11 pooled over densities (significantly for SAGE, HAN and HGT+AG at $\delta = 1$ and for HGT+AG at $\delta = 2$).

Table S4 Ring AP on unseen test rings at ring volume 4 claims per month when ring claims are also shifted ($\delta = 1$ and 2; mean over six worlds, seeds 1–6, hyperparameters of the main experiment). Chance level 0.021. The $\delta = 0$ results are in Table 4 of the article. Bold: best per column.

	$\delta = 1$			$\delta = 2$		
Method / s	0.1	0.5	0.9	0.1	0.5	0.9
GB-C	0.152	0.139	0.160	0.648	0.664	0.642
MLP-C	0.161	0.164	0.161	0.656	0.651	0.646
LR-CA	0.395	0.412	0.412	0.837	0.835	0.823
RF-CA	0.207	0.217	0.212	0.693	0.715	0.717
GB-CA	0.269	0.293	0.275	0.785	0.814	0.807
GB-CAG	0.286	0.582	0.576	0.780	0.859	0.874
SAGE	0.425	0.466	0.463	**0.866**	0.861	0.848
RGCN	0.391	0.397	0.440	0.850	0.836	0.815
HAN	**0.464**	0.444	0.410	0.853	0.870	0.857
Simple-HGN	0.338	0.419	0.403	0.805	0.818	0.791
HGT	0.251	0.303	0.285	0.779	0.794	0.778
SAGE+AG	0.409	**0.654**	**0.620**	0.855	**0.882**	**0.884**
HGT+AG	0.353	0.566	0.564	0.832	0.848	0.849
Overlap score	0.052	0.164	0.211	0.040	0.124	0.272

S6 Training-seed replication

Every reported run trains each model once per world, with training seed 0, and the tuning used the same seed. Because a network's initial weights depend only on the seed and the layer sizes, which are the same in every world, each reported GNN run starts from the weights with which its configuration was selected (checked for SAGE and HGT on a development and a test world; `paper/hiss_review/check_initial_weights.py`). To separate initialisation from world variance, GB-CAG, SAGE, HGT and the two hybrids were retrained with training seeds 1 and 2 at $\delta = 0$, all three densities and seeds 1–6, with the main hyperparameters, as fixed in the plan for the third round. Table S5 compares each method with GB-CAG with training seed 0 only and with each world's mean over the three seeds; per-density results are in the repository (`results/tables/modelseeds_comparisons.csv`).

Averaging over training seeds left the pure GNNs below GB-CAG (SAGE −0.011, HGT −0.040 pooled over densities) and reduced the hybrids' advantage (SAGE+AG +0.013 with seed 0, +0.009 with the mean of three; HGT+AG +0.012 and +0.007); with six seeds none of these pooled differences is significant after Holm adjustment.

For sparse rings, where the hybrids gained most, SAGE+AG exceeded GB-CAG by +0.043 with seed 0 (Holm $p = 0.032$) but by +0.030 with the mean of three ($p = 0.18$): its mean ring AP was 0.973 with seed 0 against 0.965 and 0.956 with seeds 1 and 2, while GB-CAG's ranged from 0.930 to 0.938. On isolated fraud the GNNs' and hybrids' deficits were unchanged (−0.086 to −0.219 with the mean of three seeds, Holm $p \leq 0.003$). Training-seed variation was small for GB-CAG (standard deviation of ring AP within a world 0.001–0.005) but, for HGT, comparable to the variation between worlds (0.022–0.034 against 0.028–0.029). The hybrids' small advantage reported in the article is therefore of the order of the variation between training seeds, and part of it may be specific to the seed used for tuning.

Table S5 Training-seed replication at $\delta = 0$ (seeds 1–6, three densities, training seeds 0–2). Ring AP difference from GB-CAG pooled over densities (seed as unit, Holm-adjusted p), with the reported training seed only and with each world's mean over three training seeds; and the standard deviation of ring AP across training seeds within a world and across worlds (means over densities).

	Seed 0 only		Mean of 3 seeds		s.d. of ring AP	
Method	Δ [95% CI]	p	Δ [95% CI]	p	training	worlds
GB-CAG	–	–	–	–	0.003	0.014
SAGE	−0.008 [−0.022, +0.005]	0.227	−0.010 [−0.026, +0.005]	0.283	0.011	0.016
HGT	−0.032 [−0.068, +0.005]	0.227	−0.040 [−0.068, −0.012]	0.060	0.027	0.028
SAGE+AG	+0.013 [+0.003, +0.023]	0.074	+0.009 [−0.002, +0.019]	0.246	0.008	0.007
HGT+AG	+0.012 [−0.004, +0.028]	0.227	+0.007 [−0.005, +0.019]	0.283	0.009	0.015

S7 Per-regime tuning

All methods were tuned once, on development worlds at $\delta = 0.5$, $s = 0.5$. To test whether the regime-level conclusions depend on that choice, the selection (the same ten candidate configurations per method, development seeds 901 and 902, validation AP; hybrids inherit their base configuration) was repeated on development worlds of the two $\delta = 0$ regimes in which the comparisons matter most, sparse ($s = 0.1$) and dense ($s = 0.9$) rings, and every method was evaluated on seeds 1–6 of that regime with the re-selected configurations, as fixed in the plan for the third round. Table S6 shows the result.

Re-tuning selected a different configuration for 8 of the 11 tuned methods at $s = 0.1$ and 7 at $s = 0.9$, but changed no method's ring AP significantly: the largest change was −0.042 (SAGE at $s = 0.9$, Holm $p = 0.51$), and methods that kept their configuration reproduced their results exactly. With every method re-tuned, no GNN or hybrid exceeded GB-CAG significantly in either regime (largest: SAGE+AG, +0.043 at $s = 0.1$, Holm $p = 0.092$). At $s = 0.9$ the pure GNNs were 0.03–0.07 below GB-CAG (significantly for Simple-HGN, Holm $p = 0.026$) and the hybrids matched it. The regime-level conclusions therefore do not depend on the regime used for tuning.

Table S6 Per-regime tuning at $\delta = 0$ (seeds 1–6): ring AP with the main configurations (tuned at $\delta = 0.5$, $s = 0.5$) and with every method re-tuned in the evaluation regime; Δ: difference from GB-CAG when all methods are re-tuned (paired over worlds, Holm-adjusted p over methods; confidence intervals in `results/tables/regimetuned_comparisons.csv`). Hybrids inherit their base configuration.

	$s = 0.1$				$s = 0.9$			
Method	Main	Re-tuned	Δ	p	Main	Re-tuned	Δ	p
LR-CA	0.923	0.923	−0.007	1.000	0.868	0.868	−0.122	<0.001
RF-CA	0.875	0.877	−0.053	0.024	0.844	0.856	−0.134	<0.001
GB-C	0.084	0.079	−0.851	<0.001	0.087	0.084	−0.906	<0.001
GB-CA	0.925	0.924	−0.006	1.000	0.889	0.884	−0.105	0.003
GB-CAG	0.930	0.930	–	–	0.990	0.989	–	–
MLP-C	0.088	0.088	−0.842	<0.001	0.095	0.093	−0.897	<0.001
SAGE	0.938	0.937	+0.007	1.000	0.976	0.934	−0.056	0.149
RGCN	0.933	0.945	+0.016	1.000	0.954	0.942	−0.047	0.085
HAN	0.908	0.906	−0.024	1.000	0.921	0.921	−0.068	0.085
Simple-HGN	0.915	0.907	−0.023	0.666	0.957	0.957	−0.033	0.026
HGT	0.910	0.910	−0.019	1.000	0.949	0.949	−0.041	0.385
SAGE+AG	0.973	0.973	+0.043	0.092	0.983	0.990	0.000	1.000
HGT+AG	0.964	0.964	+0.034	0.666	0.990	0.990	0.000	1.000

S8 Other ring shapes and sizes

The engineered graph features were designed knowing the default ring: four doctors and twelve patients linked as a block. As fixed in the plan for the fourth round (`paper/hiss_review/analysis_plan_round4.md`), every method was therefore also run on four other ring shapes and sizes at $\delta = 0$ (seeds 1–6, hyperparameters of the main experiment): a hub (one hub doctor linked to all twelve patients, each patient also linked to one of three spoke doctors); a distributed ring (eight doctors and 24 patients, each patient linked to two ring doctors drawn at random); and a small (2 doctors, 6 patients) and a large (8 and 24) block at $s = 0.5$. Each was run at the per-member claim volume of the default ring (full; 16 claims per month for twelve patients, scaled with ring size) and at half of it. Table S7 gives ring AP. Each GNN and hybrid is compared with GB-CAG within its column (Holm over the nine methods), and the structure gain GB-CAG − GB-CA across the eight columns; every comparison is in the repository (`results/tables/shapes_comparisons.csv`).

No pure GNN, general-purpose or fraud-specialised, had significantly higher ring AP than GB-CAG in any column. The largest point estimate, SAGE for the small block at full volume (+0.060, 95% CI $[-0.042, 0.163]$, Holm $p = 0.96$), comes with an interval that also allows a deficit. Pure GNNs were significantly below GB-CAG in 32 of the 56 method–column pairs, 26 of them for the hub and the large block.

The engineered features also captured structure they were not designed for. Graph features added significantly to GB-CA for the hub (+0.067 and +0.490 at full and half volume), the large block (+0.062 and +0.375) and, at half volume, the distributed ring (+0.131) and the small block (+0.377; Holm $p \leq 0.02$ across the eight columns). At full volume the distributed ring and the small block were detected by aggregates

alone (structure gain +0.003 and +0.020, not significant). The unsupervised overlap score shows where the closure statistic works: at full volume it reached 0.853 and 0.988 for the hub and the large block, but only 0.534 and 0.475 for the distributed ring and the small block.

The only significant advantage over GB-CAG was that of the hybrid SAGE+AG for the distributed ring (+0.014 $[0.008, 0.020]$ at full and +0.093 $[0.050, 0.136]$ at half volume, Holm $p = 0.019$ and 0.023), whose doctor–patient links are sparser than those of any block in the main grid. As for the sparse blocks there, message passing on top of the engineered features helped where those features were least informative. On isolated fraud every GNN and hybrid was again below GB-CAG in every column, by 0.04 to 0.37.

Table S7 Ring AP on unseen test rings for other ring shapes and sizes ($\delta = 0$, six worlds per column, seeds 1–6, hyperparameters of the main experiment). Hub: one hub doctor and three spoke doctors, twelve patients, each linked to the hub and one spoke; distributed: eight doctors and 24 patients, each patient linked to two ring doctors; small and large block: 2 doctors and 6 patients or 8 and 24, $s = 0.5$. Full: per-member claim volume of the default ring; half: half of it. Chance level 0.021–0.148. Bold: best per column.

	Hub		Distributed		Small block		Large block	
Method / volume	full	half	full	half	full	half	full	half
GB-C	0.089	0.044	0.150	0.076	0.050	0.028	0.146	0.078
MLP-C	0.089	0.045	0.158	0.086	0.048	0.028	0.157	0.083
LR-CA	0.881	0.503	0.927	0.604	0.832	0.353	0.893	0.583
RF-CA	0.845	0.288	0.936	0.489	0.694	0.148	0.918	0.599
GB-CA	0.912	0.348	0.957	0.566	0.849	0.221	0.931	0.566
GB-CAG	0.978	**0.838**	0.960	0.697	0.869	0.598	**0.993**	**0.941**
SAGE	0.929	0.574	0.962	0.704	**0.930**	0.376	0.975	0.847
RGCN	0.923	0.565	0.956	0.700	0.900	0.263	0.973	0.816
HAN	0.860	0.402	0.946	0.664	0.750	0.178	0.964	0.776
Simple-HGN	0.926	0.584	0.963	0.717	0.852	0.350	0.980	0.864
HGT	0.905	0.420	0.955	0.661	0.753	0.149	0.967	0.843
BWGNN	0.887	0.456	0.948	0.679	0.706	0.213	0.962	0.766
CARE-GNN	0.865	0.251	0.946	0.611	0.681	0.152	0.932	0.651
SAGE+AG	0.980	0.816	**0.974**	**0.790**	0.925	**0.662**	0.993	0.939
HGT+AG	**0.982**	0.799	0.966	0.735	0.892	0.579	0.990	0.926
Overlap score	0.853	0.536	0.534	0.301	0.475	0.258	0.988	0.862

S9 Cost of training and scoring

Median training time per world (Intel Core i7-1068NG7 laptop CPU, two threads per job with several jobs in parallel, so times are indicative) was 3 s for GB-CAG, 39 s for SAGE, 56 s for RGCN, 86 s for HAN, 126 s for HGT and 144 s for Simple-HGN (first-round runs; the fraud-specialised GNNs ran in the fourth round under a different load

and are not included); scoring a test period took at most 0.25 s for any method. A separate single-thread benchmark on an otherwise idle machine (four worlds at $\delta = 0$) measured scoring directly. For one audit period (about 4,900 claims and 11,200 nodes), building the aggregate and graph features took 0.05 s and the heterogeneous graph 0.01 s, and scoring every claim took 0.05 s with GB-CAG and 0.12 s with HGT. Scoring a single claim, as an online system would, took a median of 2.1 ms (99th percentile 3.1 ms, mostly library-call overhead) with GB-CAG on a precomputed feature row, and 10.7 ms (14.7 ms) with HGT on the claim's exact three-hop computation subgraph (about 126 nodes, 1% of the period graph; extraction 2.8 ms, forward pass 7.7 ms), which reproduces the full-graph score to within 1.2×10^{-7}. Routing would therefore cut the online GNN cost in proportion to the share of claims escalated. In batch scoring the escalated claims' subgraphs overlap: escalating 10% of the claims already touched 55% of the period graph, and scoring that union took 0.08 s against 0.12 s for the whole period. None of these figures includes keeping aggregates, graph features and degree counts up to date as claims arrive, which an online system would need and our code does not implement.

S10 Single-factor ablations of HGT

Table S8 changes one factor of the tuned HGT at a time in the same regime. Depth beyond two layers and the number of attention heads made no significant difference (four layers +0.002; one, two and eight heads −0.037 to +0.013; all Holm $p > 0.5$), so there is no sign of over-smoothing up to four layers on these graphs. Among the relations, removing the patient–claim edges cost most (−0.219, Holm $p = 0.005$); removing the claim–doctor edges cost 0.113 on rings (Holm $p = 0.085$) and 0.147 on isolated fraud ($p < 0.001$), while hospital nodes and the direct patient–doctor relation contributed nothing measurable. The node inputs separate the two tasks cleanly. With claim attributes set to zero, ring AP was unchanged (−0.012) and isolated AP fell to chance (0.036); an HGT that saw only the degree encodings and the graph reached a ring AP of 0.982 ± 0.018, as high as the full model, and detected no isolated fraud. HGT's ring detection is therefore structural—counts of who bills whom, propagated along patient–claim–doctor paths—and its claim attributes serve isolated fraud only. This is the information that the engineered shared-patient features summarise directly. At default volume these counts also carry the ring members' excess claims; Section 6.6 of the article shows that when they do not, the GNNs fall far behind gradient boosting with the engineered features. HGT also varied between worlds (standard deviations up to 0.11), so with six worlds differences smaller than about 0.05 cannot be resolved.

S11 Which inputs the trained models use

To see which inputs the trained models use, without retraining, we permuted one input group at a time across the test period, five times each, and recorded the drop in ring AP: for GB-CAG the C, A and G features and each G feature separately, and for the GNNs the claim attributes, static entity attributes and degree encodings across nodes of each type. At $\delta = 0$, GB-CAG relied on aggregates when rings were sparse (A: −0.80 at $s = 0.1$; G: −0.13) and switched to the graph features as the block became dense (G:

Table S8 HGT single-factor ablations at $\delta = 0$, $s = 0.9$ (six worlds; one factor changed at a time from the tuned configuration: 3 layers, 4 heads, hidden size 32). Ring AP (mean ± s.d.), isolated AP (mean) and paired change from the full model; Holm-adjusted p over every ablation comparison in this table and Table 5 of the article.

Variant	Ring AP	Δ ring AP	Isol. AP	Δ isol. AP
Full model	0.949 ± 0.052	–	0.455	–
Depth (full: 3)				
1 layer	0.788 ± 0.052	−0.161 ($p = 0.020$)	0.239	−0.216 ($p = 0.007$)
2 layers	0.947 ± 0.035	−0.001 ($p = 1.000$)	0.423	−0.032 ($p = 1.000$)
4 layers	0.951 ± 0.043	+0.002 ($p = 1.000$)	0.461	+0.005 ($p = 1.000$)
Heads (full: 4)				
1 attention head	0.920 ± 0.049	−0.028 ($p = 1.000$)	0.384	−0.071 ($p = 0.228$)
2 heads	0.911 ± 0.081	−0.037 ($p = 0.506$)	0.473	+0.018 ($p = 0.813$)
8 heads	0.961 ± 0.023	+0.013 ($p = 1.000$)	0.524	+0.069 ($p = 0.282$)
Relations				
No patient–claim relation	0.730 ± 0.055	−0.219 ($p = 0.005$)	0.311	−0.144 ($p = 0.333$)
No claim–doctor relation	0.836 ± 0.037	−0.113 ($p = 0.085$)	0.308	−0.147 ($p < 0.001$)
No patient–doctor relation	0.902 ± 0.055	−0.047 ($p = 0.413$)	0.423	−0.032 ($p = 1.000$)
No hospital nodes	0.946 ± 0.032	−0.002 ($p = 1.000$)	0.427	−0.028 ($p = 1.000$)
Node inputs				
No degree encodings	0.569 ± 0.110	−0.379 ($p = 0.003$)	0.375	−0.080 ($p = 0.813$)
Claim attributes zeroed	0.937 ± 0.064	−0.012 ($p = 1.000$)	0.036	−0.419 ($p < 0.001$)
Entity attributes zeroed	0.969 ± 0.027	+0.020 ($p = 1.000$)	0.444	−0.011 ($p = 1.000$)
Degree encodings only	0.982 ± 0.018	+0.033 ($p = 1.000$)	0.040	−0.415 ($p < 0.001$)

−0.81 and −0.88 at $s = 0.5$ and 0.9; A: below 0.06), above all the normalised triadic closure of the claim (−0.47 on its own at $s = 0.9$). SAGE and HGT relied almost entirely on degree encodings (−0.76 to −0.89 at every density) and hardly at all on claim or entity attributes (below 0.05). The two kinds of score are not on a common scale: GB-CAG's measure reliance on engineered summaries of the graph, whereas a GNN's degree encodings are raw counts whose relational meaning is extracted by message passing, and permutation understates groups whose information is duplicated elsewhere. Read together with the ablations, they are consistent with both model families detecting dense rings from the same evidence: how many counterparties a claim's patient and doctor have, and how many of them they share. (Full results, including isolated fraud and test-time rewiring, are in the repository.)

S12 Figures of the mechanism ablations and of new versus recurring rings

Figs. S2 and S3 show the results of Table 5 and Table 6 of the article with 95% intervals.

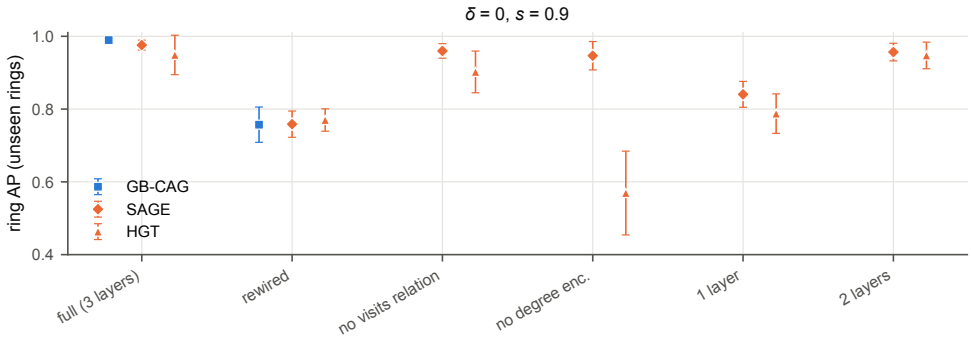


Fig. S2 Mechanism ablations at $\delta = 0$, $s = 0.9$ (mean and 95% t-interval over six worlds). Rewiring preserves every node's degree but destroys the ring block.

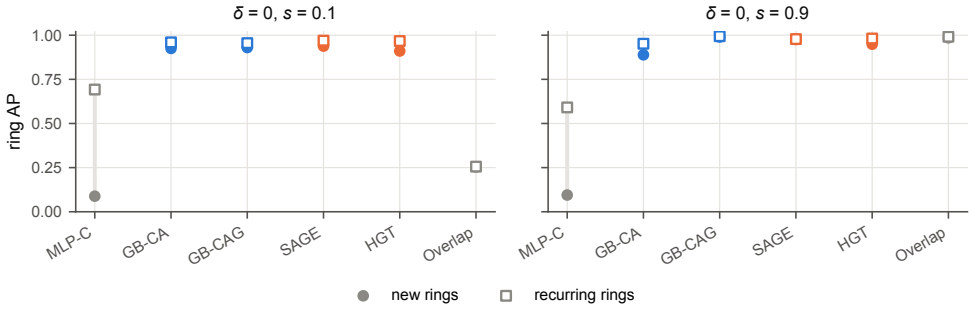


Fig. S3 Ring AP on new versus recurring rings ($\delta = 0$, six worlds).

S13 Repeated cross-validation and per-rotation tuning on the public data

Both analyses were specified in the plan for the second round of experiments before they were run.

Repeated cross-validation. The provider-disjoint five-fold rotation of the article (fold k test, fold $k + 1$ validation, three training folds) was repeated with the stratified fold assignment reshuffled four more times (seeds 1–4), giving 25 rotations with the article's split as the first. The second round covered the tabular models, the MLP and three GNNs (SAGE, HAN, HGT); RGCN, Simple-HGN and the two hybrids were added in the third round and the fraud-specialised BWGNN and CARE-GNN in the fourth, with the same reshuffled folds (`paper/hiss_review/analysis_plan_round3.md` and `round4.md`). Hyperparameters are those of the shared selection. Each method is compared with GB-P by the corrected repeated cross-validation t-test (Nadeau and Bengio 2003; Bouckaert and Frank 2004; references 52 and 53 of the article), which inflates the variance of the mean difference by $1/25 + 1/3$ instead of $1/25$, with 24 degrees of freedom, and the fourteen p-values are Holm-adjusted together (the earlier rounds adjusted eight and twelve within smaller

families). Table S9 shows the result. GB-P has the highest mean AP; graph features change it by −0.007 and −0.006, with upper 95% limits of +0.006 and +0.007; the GNNs are 0.025–0.092 lower and the hybrids 0.039–0.063 lower. After adjustment only RGCN differs significantly (−0.092, Holm $p = 0.009$); the next smallest adjusted p is 0.062 (SAGE+G). Without the correction, every difference would be significant (all unadjusted $p < 0.01$), which shows how strongly overlapping training sets inflate a plain paired test.

Table S9 Public data, five-fold rotation repeated five times (25 rotations; the first repeat is the split reported in the article): AP mean ± s.d., and the difference from GB-P with the corrected repeated cross-validation 95% CI and Holm-adjusted p (variance factor $1/25 + 1/3$, 24 degrees of freedom).

Method	AP	Δ vs. GB-P [95% CI]	p
LR-P	0.674 ± 0.046	−0.019 [−0.044, +0.006]	0.905
RF-P	0.674 ± 0.041	−0.019 [−0.042, +0.005]	0.905
GB-P	0.693 ± 0.045	–	–
GB-PG	0.686 ± 0.045	−0.007 [−0.019, +0.006]	0.905
GB-PGL	0.687 ± 0.046	−0.006 [−0.019, +0.007]	0.905
MLP-P	0.676 ± 0.042	−0.017 [−0.042, +0.009]	0.905
SAGE	0.665 ± 0.058	−0.028 [−0.066, +0.010]	0.905
RGCN	0.601 ± 0.051	−0.092 [−0.140, −0.043]	0.009
HAN	0.659 ± 0.046	−0.034 [−0.066, −0.002]	0.398
Simple-HGN	0.666 ± 0.048	−0.027 [−0.063, +0.008]	0.905
HGT	0.668 ± 0.050	−0.025 [−0.055, +0.005]	0.905
BWGNN	0.665 ± 0.051	−0.028 [−0.065, +0.008]	0.905
CARE-GNN	0.661 ± 0.043	−0.032 [−0.065, 0.000]	0.500
SAGE+G	0.630 ± 0.049	−0.063 [−0.105, −0.021]	0.062
HGT+G	0.654 ± 0.041	−0.039 [−0.068, −0.009]	0.146

Per-rotation (nested) selection. The shared selection uses a separate 60/20/20 split of the same providers: 18–22% of the providers of each test fold of the first repetition were in its validation set and 59–62% in its training set. The 25 rotations above all use these shared configurations. Re-selecting each method's configuration inside every rotation, from the same six candidates and on that rotation's validation fold only, removes this overlap. It was done for the first repetition only. Table S10 compares nested with shared selection, and every method with GB-P when both are selected inside each rotation. Re-selection changed AP by at most 0.010 for the tabular models and the MLP and by −0.008 (HGT) to +0.036 (RGCN) for the GNNs and hybrids, with no change significant; SAGE selected its shared configuration in all five rotations, and SAGE+G inherits it, so their results are identical. We had expected the shared selection to favour the GNNs; it did not do so systematically. With every method selected inside each rotation, GB-P remains the best method, and every GNN and hybrid is 0.023–0.053 below it (smallest Holm $p = 0.059$ for SAGE and SAGE+G).

Table S10 Public data: hyperparameters re-selected inside every rotation (nested) versus selected once on a separate split (shared; Table 7 of the article). AP mean over the five rotations. Nested − shared: change from re-selection; vs. nested GB-P: difference from GB-P when both are selected inside each rotation, with Holm-adjusted p. Intervals are corrected resampled 95% CIs. Same choice: the shared configuration was selected in every rotation, so the results are identical. Hybrids inherit their base model's per-rotation choice.

Method	Nested AP	Nested − shared [95% CI]	vs. nested GB-P [95% CI]	p
LR-P	0.682	+0.003 [−0.008, +0.015]	−0.024 [−0.055, +0.007]	0.708
RF-P	0.680	−0.001 [−0.021, +0.019]	−0.026 [−0.062, +0.010]	0.708
GB-P	0.706	+0.002 [−0.016, +0.019]	–	–
GB-PG	0.693	−0.001 [−0.033, +0.030]	−0.012 [−0.045, +0.020]	0.708
GB-PGL	0.687	−0.010 [−0.040, +0.020]	−0.019 [−0.057, +0.018]	0.708
MLP-P	0.688	+0.003 [−0.008, +0.014]	−0.018 [−0.045, +0.010]	0.708
SAGE	0.664	same choice	−0.042 [−0.062, −0.022]	0.059
RGCN	0.657	+0.036 [−0.014, +0.086]	−0.049 [−0.103, +0.005]	0.585
HAN	0.670	+0.003 [−0.023, +0.030]	−0.036 [−0.089, +0.017]	0.708
Simple-HGN	0.682	+0.012 [−0.032, +0.056]	−0.023 [−0.048, +0.001]	0.585
HGT	0.668	−0.008 [−0.034, +0.019]	−0.038 [−0.065, −0.011]	0.213
BWGNN	0.670	−0.003 [−0.018, +0.012]	−0.036 [−0.100, +0.029]	0.708
CARE-GNN	0.673	+0.005 [−0.019, +0.028]	−0.032 [−0.060, −0.005]	0.352
SAGE+G	0.653	same choice	−0.053 [−0.078, −0.027]	0.059
HGT+G	0.661	+0.013 [−0.018, +0.043]	−0.045 [−0.096, +0.006]	0.585

S14 Re-analysis of a small fixed benchmark

To show how a small, fixed benchmark can mislead, we re-analysed one: a synthetic dataset of 2,100 claims with five injected rings, produced by an earlier, simpler generator of ours that is included in the repository, evaluated with its own date-sorted 60/20/20 split (Table S11). Such a dataset appears to test ring detection, but this one cannot tell whether relational structure helps, for three reasons. First, its ring claims were drawn from the same inflated distribution as its isolated fraud, so they were anomalous on their own: claim amount alone separates ring from legitimate claims with ROC-AUC 0.94, more than in our most anomalous regime ($\delta = 2$, 0.90), and gradient boosting on three raw claim attributes ranks all 21 test ring claims above every legitimate claim (ring AP 1.000). Second, the same five rings were active for all 24 months, so every test ring claim was billed by a doctor with labelled ring claims in training, which is the memorisation setting of Section 7 of the article. Third, the test split held 21 ring claims (3–6 per ring): one claim moves ring recall by 4.8 points, even a perfect recall has an exact 95% interval of $[0.84, 1.00]$, so a recall gap of a few claims between two models is within noise, and a single dataset says nothing about variation between datasets. None of these properties is peculiar to that dataset. A small, fixed benchmark whose injected rings differ from legitimate claims in their features, recur across the split and contribute a few dozen test claims cannot separate relational learning from claim-level anomaly detection or from recognising known members. Controlling the feature-level signal, testing on new rings and replicating worlds removes all three confounds. Because the generator, features, evaluation unit and tuning differ, numbers from the two set-ups are not directly comparable.

Table S11 A small fixed benchmark (2,100 claims, five rings; an earlier, simpler generator of ours, included in the repository), re-analysed with its own split (date-sorted, 60/20/20), compared with our benchmark. Benchmark ranges span $\delta = 0$ to $\delta = 2$ ($s = 0.5$, six worlds each; ring AP and recall of GB-C pooled over s). Shift and ROC-AUC use raw claim amounts, not adjusted for specialty, for both. Interval: exact 95% binomial.

	Fixed dataset	This benchmark
Claims in total (in the test split / period)	2,100 (420)	20,176 (4,924), mean per world
Ring claims in test	21 (3–6 per ring)	384 (4 rings × 96)
Rings	5, active all 24 months	4 per period, each active 6 months
Test-ring doctors with labelled ring claims in training	100%	0% (by design)
Ring vs. legitimate log-amount shift (s.d.)	1.49	0.11 ($\delta = 0$) to 1.87 ($\delta = 2$)
ROC-AUC of claim amount alone, ring vs. legitimate	0.94	0.53 to 0.90
Ring AP, GB on claim attributes only	1.000	0.086 to 0.811
Ring recall, GB on claim attributes only	21/21 [0.84, 1.00]	0.15 to 0.76
Generated datasets (independent generator seeds)	1 (1)	54 (6)

S15 Wide hyperparameter search

A plan committed before the runs (`paper/hiss_review/analysis_plan_round5.md`) searched more widely for SAGE, RGCN, Simple-HGN and HGT, and gave GB-CAG the same budget. Each method received 30 candidates: its main configuration and 29 draws (sampling seed 1) from a wider grid, none of which any earlier tuning had offered. For the GNNs the grid covered hidden size 16–128, 2–4 layers, dropout 0–0.3, learning rate 3×10^{-4} to 10^{-2}, weight decay 10^{-5} to 10^{-3}, and sum or mean aggregation or 2–8 attention heads. For GB-CAG it covered learning rate 0.01–0.2, 7–127 leaves, minimum leaf size 5–50, ℓ_2 penalty 0–10 and 200–800 iterations. Candidates were selected and early-stopped on validation ring AP, over at most 300 epochs with patience 30; the main protocol uses overall AP, 200 and 20. All 30 ran on development world 901, drawn at the evaluation regime ($\delta = 0$ and the cell's volume and s), and the six best also on world 902. The best mean over the two was chosen. Evaluation used seeds 1–6 and training seed 0, and the training label was unchanged.

The selected configuration replaced the main one in every case except Simple-HGN at volume 4 and $s = 0.5$ (Table S12). The plan's rule for volume 4 is that the claim that pure GNNs trail GB-CAG stands for a cell only if every GNN's mean stays below GB-CAG's and the upper 95% limit of its difference is below zero. Both cells meet it: the largest upper limit is −0.109. At default volume no GNN was above GB-CAG, so the statement that no pure GNN outperformed it needs no qualification.

The wider search changed little. Only SAGE at volume 4 and $s = 0.9$ improved significantly (+0.063, 95% CI $[0.028, 0.098]$, Holm $p = 0.029$). GB-CAG's ring AP

changed by at most 0.022. HGT's selected configuration at volume 4 and $s = 0.9$ (hidden size 128, two layers, eight heads, learning rate 3×10^{-4}) had a validation ring AP of 0.19 on the development worlds but reached 0.029 on the evaluation worlds. Its best validation epoch came within the first five in five of six worlds, so early stopping ended training before it learned. Its main configuration reached 0.098. We report the failure as it occurred; it does not affect the decision, because both configurations are far below GB-CAG. Validation ring AP varied widely between candidates on the low-volume development worlds (0.02 to 0.27 on the first world), so selection from two worlds is noisy. That is the main limit of this search.

Table S12 Wide hyperparameter search (ring AP, $\delta = 0$, six worlds, seeds 1–6). Main: the configurations of the main experiment. Wide: 30 candidates from a wider grid (the main configuration and 29 not offered before), selected and early-stopped on validation ring AP on development worlds of the evaluation regime. Wide − main: paired change. Δ: difference from GB-CAG when both are searched widely, with Holm-adjusted p over the four GNNs of the cell.

Method	Main	Wide	Wide − main [95% CI]	Δ [95% CI]	p
Volume 4, $s = 0.5$					
GB-CAG	0.409	0.407	−0.001 [−0.040, +0.037]	–	–
SAGE	0.108	0.146	+0.038 [−0.030, +0.106]	−0.261 [−0.414, −0.109]	0.013
RGCN	0.112	0.124	+0.012 [−0.017, +0.041]	−0.283 [−0.404, −0.161]	0.008
Simple-HGN	0.133	0.137	+0.004 [−0.038, +0.046]	−0.271 [−0.427, −0.114]	0.013
HGT	0.079	0.090	+0.011 [−0.016, +0.039]	−0.317 [−0.455, −0.179]	0.008
Volume 4, $s = 0.9$					
GB-CAG	0.570	0.593	+0.022 [−0.001, +0.045]	–	–
SAGE	0.131	0.193	+0.063 [+0.028, +0.098]	−0.399 [−0.468, −0.331]	<0.001
RGCN	0.142	0.169	+0.027 [−0.055, +0.109]	−0.424 [−0.482, −0.366]	<0.001
Simple-HGN	0.151	0.206	+0.055 [−0.001, +0.110]	−0.387 [−0.461, −0.313]	<0.001
HGT	0.098	0.029	−0.069 [−0.148, +0.009]	−0.564 [−0.662, −0.465]	<0.001
Volume 16, $s = 0.9$					
GB-CAG	0.990	0.991	+0.002 [0.000, +0.003]	–	–
SAGE	0.976	0.984	+0.008 [−0.001, +0.017]	−0.008 [−0.015, 0.000]	0.089
RGCN	0.954	0.969	+0.015 [−0.005, +0.036]	−0.022 [−0.035, −0.009]	0.028
Simple-HGN	0.957	0.961	+0.004 [−0.014, +0.021]	−0.031 [−0.049, −0.012]	0.028
HGT	0.949	0.928	−0.020 [−0.042, +0.001]	−0.063 [−0.134, +0.008]	0.089

S16 Closure columns

For a claim of patient p and doctor d, the closure of the G features is $\sum_{d' \in N(p) \setminus \{d\}} D_{d'd}$, where $N(p)$ is the patient's doctors and $D_{d'd}$ the number of patients shared by doctors d' and d in the period. Each term counts p once, and every other shared patient p' closes a four-cycle p–d–p'–d'–p through the claim's edge. The closure therefore equals $\deg(p) - 1$ plus the number of those four-cycles, which we confirmed on 800 sampled claims of a generated world. Message-passing networks cannot perform induced-subgraph counting of any connected pattern of three or more

nodes (reference 37 of the article), and a four-cycle in a bipartite graph is necessarily induced. A small example shows the limit directly: an eight-cycle and two disjoint four-cycles are both bipartite and two-regular, so with identical node attributes message passing gives every node the same computation tree, although only the second graph contains four-cycles. The limit is a worst-case one, and real node attributes can break such ties.

A plan committed before the runs (`paper/hiss_review/analysis_plan_round5.md`) tested whether this is what the pure GNNs lack. At $\delta = 0$, volumes 8 and 4 and $s = 0.5$ and 0.9 (seeds 1–6, hyperparameters of the main experiment), it added the two closure columns, raw and degree-normalised, to the claim nodes of SAGE and HGT (SAGE+Cl, HGT+Cl) and to GB-CA (GB-CA+Cl), and removed them from GB-CAG, keeping its other five graph features (GB-CAG−Cl). It predicted that SAGE+Cl and HGT+Cl would recover at least half of the gap between the pure GNN and GB-CAG, and that GB-CAG−Cl would lose most of the structure gain.

The first part held and the second did not (Table S13). The columns recovered 82–90% of SAGE's gap and 52–84% of HGT's, with gains of 0.15–0.36 (Holm $p \leq 0.027$ over the four cells of each comparison). SAGE+Cl stayed 0.02–0.08 below GB-CAG (Holm $p \geq 0.18$), and HGT+Cl 0.16–0.22 below it at volume 4 (Holm $p \leq 0.033$) and 0.04–0.08 below it at volume 8 (Holm $p = 0.061$). With six seeds these intervals are wide and do not show equivalence: at volume 4 and $s = 0.9$ SAGE+Cl differed from GB-CAG by -0.080 (95% CI $[-0.224, 0.064]$). GB-CA+Cl reached 30–90% of the structure gain. GB-CAG−Cl lost 0.01–0.07 (Holm $p \geq 0.21$) and kept 79–98% of the structure gain, so the second part of the prediction failed.

The plan's reporting rule for a failed prediction says that the article does not offer four-cycle counting as the explanation and that it states that the tested GNNs cannot count such patterns in general. We follow it. The rule also anticipated that the columns would not close the gap, which is not what happened, so we report instead what the columns did. After seeing the result we noted that the prediction could not have isolated the closure: of the five remaining columns, the number of doctors sharing at least two patients with the claim's doctor and the patient's share of such doctor pairs both test for a second shared patient, which is itself a four-cycle, and the two Jaccard columns derive from the same shared-patient counts. Only the mean degree of the doctor's patients does not. This reading is post hoc, and no run tested it, because the plan excluded additions after the results were seen.

Table S13 Which engineered columns matter (ring AP, $\delta = 0$, six worlds, seeds 1–6, main hyperparameters). +Cl: the two closure columns added; −Cl: the two closure columns removed, the other five graph features kept.

Volume, s	GB-CA	GB-CA+Cl	GB-CAG−Cl	GB-CAG	SAGE	SAGE+Cl	HGT	HGT+Cl
8, 0.5	0.372	0.795	0.864	0.879	0.683	0.856	0.649	0.802
8, 0.9	0.311	0.864	0.912	0.923	0.755	0.906	0.689	0.885
4, 0.5	0.052	0.160	0.334	0.409	0.108	0.363	0.079	0.251
4, 0.9	0.055	0.331	0.525	0.570	0.131	0.491	0.098	0.348

S17 Group-fraud flags on the public data

Ma et al. (reference 13 of the article) added expert-annotated group-fraud flags to the public dataset used in the article, and published them at `https://github.com/finint/RDPGL` (commit `6e0151d`). Anti-fraud experts of China's National Healthcare Security Administration judged the records from domain expertise; the criterion is not documented, and neither group identifiers nor group sizes are published. The flags cover 100 of the 506 fraudulent providers and 109,683 of the 558,211 claims. They are constant within a provider and never mark a provider that is not labelled fraudulent. The release recodes identifiers. We mapped them to the original ones by rank and confirmed the mapping for all 5,410 providers on claim count, inpatient count, total reimbursement, total deductible and fraud label.

A plan committed before any score was computed (`paper/hiss_review/analysis_plan_round6.md`) fixed the analysis below. The flagged providers were already known to hold about half of the fraud providers' claims, so the map of the article made two predictions: graph features would add little (P1), and no pure GNN would be significantly above gradient boosting with graph features (P2).

Stored predictions.

We re-scored the stored test predictions of all 15 methods on all 25 rotations; nothing was retrained. All 375 sets of predictions first reproduced their recorded AP. Models had been trained on the provider fraud label, as synthetic models are trained on any fraud and scored on rings. Group AP ranks flagged providers against non-fraud providers with the other fraud providers excluded, and individual AP ranks the other fraud providers against non-fraud providers (Table S14). Tests are the corrected repeated cross-validation tests of Section S13, with Holm adjustment within each family.

Both predictions held. Graph features added nothing to GB-P (GB-PG -0.001, 95% CI $[-0.026, 0.024]$; GB-PGL 0.000 $[-0.035, 0.035]$; Holm $p = 1.000$). All nine graph models were below GB-PG, by 0.10–0.21 (SAGE+G significantly, Holm $p = 0.031$; the others Holm $p \geq 0.090$). The structure gain was no larger on group AP than on individual AP (difference $+0.010$ $[-0.021, 0.040]$, $p = 0.510$).

Retraining on the flag.

Those models were not trained to find groups, so GB-P and GB-PG were retrained on the flag on all 25 rotations, and SAGE, RGCN and HGT on the five rotations of the first repetition, with the other fraud providers removed from every split and hyperparameters unchanged. GB-P reached 0.764 and GB-PG 0.755 (difference -0.009 $[-0.072, 0.055]$, $p = 0.785$). Over the five rotations GB-PG reached 0.775, SAGE 0.711 (-0.065 $[-0.235, 0.105]$), RGCN 0.640 (-0.136 $[-0.456, 0.185]$) and HGT 0.758 (-0.018 $[-0.256, 0.221]$), with Holm $p = 0.916$ for each. Training on the flag narrowed HGT's gap to GB-PG, from -0.099 on all 25 rotations of the stored predictions. With five rotations these tests have little power, and the intervals are compatible with sizeable differences in either direction.

Description of the flags.

Flagged providers are large: a median of 618 claims, against 114.5 for other fraud providers and 27 for non-fraud providers. Through shared beneficiaries, which correspond to the benchmark's ring structure, they are no more densely linked than other fraud providers: median maximum Jaccard overlap 0.087 and 0.088, and share of beneficiaries shared with another provider 0.895 and 0.894. They do have more co-providers, as large providers do. Assortativity through shared beneficiaries is weak: on average 11.7% of a flagged provider's co-providers are flagged, against 10.1% for unflagged providers and a base rate of 1.8%, which is high for both because shared beneficiaries concentrate on large providers. Through shared physicians it is marked: 13.7% against 2.9%. We cannot tell whether this reflects coordination, the size of these providers or the annotation criterion, and it did not help detection, because aggregates already rank these providers highly and the physician features are among the inputs of GB-PG.

Reading.

By the plan's rules the result is called consistent with the map, not a validation. The flags are an undocumented expert judgement on claims that may be synthetic (Section 5.1 of the article). If the experts used shared links, any relational signal would be partly built in. On the relation that corresponds to the benchmark's rings the flagged providers are not denser, so these data sit in the regime of unusually active members and do not test dense rings of otherwise unremarkable members.

Ma et al. report that their RDPGL exceeded baselines on these flags. Its scores are not comparable with ours. It scores claims on a temporal split, and at least 4,058 of its 4,512 outpatient test-period providers also appear in training (from its Table 1), whereas ours are disjoint providers. Its code was not available at the commit we used (the README says "Coming soon"), so RDPGL itself was not run.

Table S14 Public data re-scored on the published group-fraud flags (25 rotations, models trained on the provider fraud label, nothing retrained). Group AP: 100 flagged providers against non-fraud providers, other fraud providers excluded (chance 0.020). Individual AP: the other fraud providers against non-fraud providers (chance 0.076). Mean ± s.d. over rotations. Δ: difference in group AP from GB-PG (or, for GB-PG and GB-PGL, from GB-P), with the corrected repeated cross-validation 95% CI and Holm-adjusted p within each family.

Method	Group AP	Individual AP	Δ [95% CI]	p
LR-P	0.677 ± 0.113	0.572 ± 0.049	–	–
RF-P	0.626 ± 0.105	0.580 ± 0.047	–	–
GB-P	0.719 ± 0.109	0.596 ± 0.050	–	–
GB-PG	0.719 ± 0.119	0.585 ± 0.050	-0.001 $[-0.026, 0.024]$	1.000
GB-PGL	0.719 ± 0.117	0.587 ± 0.053	$+0.000$ $[-0.035, 0.035]$	1.000
MLP-P	0.656 ± 0.105	0.581 ± 0.044	–	–
SAGE	0.617 ± 0.158	0.574 ± 0.055	-0.102 $[-0.212, 0.008]$	0.182
RGCN	0.531 ± 0.120	0.499 ± 0.056	-0.187 $[-0.328, -0.046]$	0.091
HAN	0.588 ± 0.111	0.571 ± 0.050	-0.130 $[-0.239, -0.022]$	0.122
Simple-HGN	0.610 ± 0.124	0.572 ± 0.053	-0.109 $[-0.216, -0.002]$	0.182
HGT	0.620 ± 0.111	0.577 ± 0.056	-0.099 $[-0.197, -0.000]$	0.182
BWGNN	0.623 ± 0.122	0.571 ± 0.053	-0.095 $[-0.225, 0.035]$	0.182
CARE-GNN	0.591 ± 0.112	0.569 ± 0.042	-0.128 $[-0.239, -0.017]$	0.130
SAGE+G	0.504 ± 0.137	0.543 ± 0.052	-0.215 $[-0.351, -0.078]$	0.031
HGT+G	0.602 ± 0.111	0.559 ± 0.049	-0.116 $[-0.209, -0.023]$	0.116